Magnetic disturbance and noise: what filtering is expected to handle and what it is not

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Abstract

Geomagnetic positioning uses measurements made within the magnetic surroundings of the equipment carrying the sensors. Motors, steel structures, electrical loads and nearby activity belong in the assessment of those measurements, alongside natural field variation and the condition of the sensors themselves. NSATS's geomagnetic technology determines position from Earth's magnetic field without reliance on live GPS satellite signals. NSATS's proprietary noise filtering processes the sensor measurements used for that calculation. Its expected role in addressing platform and short-period noise must be assessed under stated conditions. A changed magnetic environment and a stationary sensor beside cycling machinery require explicit evaluation. This paper sets out the disturbance classes to record, the distinction between field measurements and positioning evidence, and the installation decisions that an organisation needs to examine before accepting its own implementation.

Classes of disturbance

The magnetic surroundings of a positioning device include the platform carrying it and the environment through which it moves. The organisation planning an implementation needs to identify both. A description such as indoor, underground or airborne names an intended application environment; it does not identify which electrical loads were active or what magnetic material surrounded the sensors during an observation.

Table 1 provides the disturbance classes for that inventory. Its entries are conditions to examine, with no implied ranking of severity or statement that a particular condition has been handled by NSATS's own research and development (R&D) demonstration arrangement.

Table 1. Disturbance classes and conditions for an implementation's evaluation (conditions to record, not measurements).

Disturbance class	Conditions
Platform-borne	motors, alternators, steel structure, electrical loads

Disturbance class	Conditions
Environmental	ferrous buildings, vehicles passing, power lines, rail
Geophysical	diurnal variation, geomagnetic storms
Sensor-related	temperature, mounting, calibration drift

Platform and environmental conditions

Platform-borne sources travel with the sensors. The equipment record should identify the motors, alternators, steel structure and electrical loads relevant to the chosen mounting. Recording the platform's operating state connects the magnetic observations to the work being performed. A record that names a vehicle but omits whether its equipment was running leaves part of the measurement conditions unspecified.

Environmental sources belong to the surroundings. Ferrous buildings, passing vehicles, power lines and rail are included in the inventory because a platform can encounter them during its task. Fixed structures and moving objects need separate descriptions. The former describe the surroundings along the route; the latter introduce activity that may differ between visits. These distinctions follow the site's account of local magnetic surroundings.

The categories organise the observation record without assigning every field change to a cause. Machinery can be part of the platform in one implementation and part of the surroundings in another. Its relationship to the sensors and its operating state determine how it should be recorded. If equipment activity and a field change occur together, the record establishes their timing; attributing the change to that equipment requires an assessment that supports the connection.

Natural field variation

Earth's main magnetic field changes over time. The International Geomagnetic Reference Field (IGRF), maintained by the International Association of Geomagnetism and Aeronomy, describes that main field and its secular variation, the term used for its slow change. 1 It provides scientific background for understanding geomagnetic measurements; it is not a record of the magnetic conditions beside a particular device.

Diurnal variation means variation over the day. Natural magnetic observations also include irregular disturbances. The British Geological Survey's account of Earth's field distinguishes the regular daily variation from disturbed conditions, including magnetic storms. 2 These changes belong in the geophysical class because the field being measured can change while the sensor remains in place.

A geomagnetic storm is a disturbance of Earth's magnetosphere associated with energy transferred from the solar wind. 3 Space-weather services supply context for planning and interpreting magnetic observations. That context does not establish the response of NSATS's arrangement to a storm; the positioning evidence must come from recorded work under identified conditions.

Sensor conditions

Temperature, mounting and calibration drift concern the measuring equipment. They need their own entries even when the platform and surrounding structures are unchanged. The sensor manufacturer's instructions govern any required calibration; the installation record identifies the mounting and the checks performed. The distinction matters because an observation depends on the instrument used as well as the field present around it.

For technical readers: the table is an inventory of conditions, not a diagnosis derived from a field value. A raw measurement does not carry a label identifying which listed source contributed to it. Record sensor conditions and surrounding activity independently of the calculated position so that the assessment retains the context needed to interpret both.

What filtering is expected to address

NSATS's geomagnetic positioning uses sensors on the platform to measure Earth's magnetic field. NSATS's proprietary noise filtering processes those measurements, and the arrangement calculates position from the filtered values.

The expectation to examine is the handling of platform and short-period noise. "Platform" identifies the equipment carrying the sensors. "Short-period" identifies variation over a short interval, without specifying a duration or a processing boundary. Neither term supplies an acceptance criterion by itself. An organisation needs to identify the equipment state, movement and surroundings represented by its evaluation and assess the resulting position against an independent reference.

A source inventory makes that expectation concrete. For platform noise, the evaluation plan can identify the relevant electrical loads and machinery states. For changing surroundings, it can identify the nearby activity that the intended task encounters. The assessment then concerns position output under those recorded conditions. No performance figure follows from placing a source in either category.

A changed environment

A change in the field can be a change in the environment itself. Moving ferrous equipment or changing electrical activity alters the conditions in which measurements are made. The implementation plan must therefore allow for the surroundings present during use, including changes after installation. It cannot assume that naming noise filtering establishes operation through every altered magnetic environment.

This distinction also applies to a fixed structure encountered along a route. The inventory records its presence as part of that route's conditions; it does not state what the processing does with that structure's contribution. The practical requirement is that the evaluation includes the surroundings relevant to the task and keeps the observations associated with them.

A stationary sensor beside cycling machinery

A sensor remaining in place beside machinery that changes operating state presents a specific condition for evaluation. The sensor's position can remain fixed while its magnetic surroundings change. A stationary check should record that distinction explicitly, with a reference appropriate to the

fixed position and an account of the machinery's activity.

The expectation for platform and short-period noise does not establish the positioning response in this condition. Treating every changing measurement as removable noise would leave the changed surroundings unexamined. The stationary condition needs its own observation record, particularly where a task alternates between travel and work at a fixed point. A moving record cannot answer every question about stationary use.

Position output and application behaviour

Field magnitude and position answer different questions. A field measurement describes the magnetic input; a calculated position supplies information that the application can use. Assessing that position requires an independent reference for the same observation event. A change between raw and filtered values does not, by itself, establish closer agreement between calculated and reference positions. Positioning from Earth's magnetic field explains why an assessment needs the reference and conditions as well as the reported position.

The surrounding application also needs defined behaviour when the available position information cannot support its task. Security and positioning resilience discusses behaviour under defined disturbances, and interruption and resumption, in terms of the application's requirements. Such behaviour belongs in the implementation plan; it is not an additional function inferred from noise filtering. Deliberate magnetic interference belongs within the separate security evaluation scope. Positioning validation data is not security evidence.

How a run records disturbances

NSATS's own R&D demonstration arrangement uses eight sensors. Its recording format associates the measured field before and after NSATS's proprietary noise filtering with each sensor and point. It also records the calculated position and the corresponding reference position. This allows the published record, when approved, to keep the magnetic measurements and the positioning assessment attached to the same observation.

Field columns and conditions

The `field_raw_nT` column holds measured geomagnetic field magnitude before processing. The `field_filtered_nT` column holds the magnitude after processing. Both use nanotesla. The `sensor_id`, `point_index` and elapsed-seconds field `t_s` identify the sensor and observation. Reading those identifiers together avoids treating values from different observations as a before-and-after pair.

The run description supplies the setting, movement case, platform, reference type and software version. Installation and equipment-state information supplies context for interpreting those fields. A reviewer should be able to identify the configuration represented by an observation and any recorded change during the run. Otherwise a difference in observations may be discussed without knowing whether the measurement conditions stayed the same.

The `notes` column is internal. It can retain the disturbance context required for review, but its contents are not automatically public text. A releasable account must respect the record's publication boundaries. In particular, a description of nearby equipment must not disclose a place, facility or

absolute position. A coarse setting and an approved description of conditions provide context without identifying the operating site.

The stationary check

For a stationary check, the record needs to establish that the reference position remained fixed and identify the equipment states observed during the check. The field columns retain the raw and filtered observations for each sensor. Internal notes can connect those observations with cycling machinery or other surrounding activity. An approved record establishes what was done; without one, the check is a planned assessment condition, not an observation.

This paper reports no approved stationary-check record. The sentences above describe what such a record would need to establish, not an observation that exists.

A run record has no dedicated stationary-check column; its field, reference and notes entries are where the observation and its context are recorded. They do not supply an acceptance threshold. That threshold belongs to the organisation's requirements for the task, agreed before the observations are interpreted.

Reading position deviation

At each point the arrangement calculates a position afresh. Deviation is the horizontal distance between the calculated position and the reference position at that point. As a percentage, it is that distance divided by the distance moved since the previous point, multiplied by `100`. The start point has no previous point and is excluded.

For technical readers: a stationary interval has no reference movement to use as the percentage denominator. Its percentage is undefined; it must not be presented as zero or assigned a substitute denominator. The distance deviation remains meaningful. The same issue requires attention wherever a movement record includes a stop, so the assessment should identify those intervals when reading the percentage field.

The reference has its own uncertainty and conditions of use. Those belong with the positioning observations, as explained in How positioning is evaluated. Keeping the reference independent of the calculated output allows the assessment to examine agreement without using the output to certify itself. The field measurements provide context for that assessment; they do not replace the reference.

What is published

Validation results is where recorded runs of NSATS's own R&D demonstration arrangement appear, each once it has been approved. Measured field values and position deviations belong in those validation records; this paper contains no numerical disturbance result. The related accounts of recorded work belong on Geomagnetism demonstration records.

A public run is identified by its run ID and its start-point code, `gps_loc_code`. Positions are expressed as metres north and east of the start. Start time and elapsed seconds identify timing within the run; run dates and absolute locations are withheld. A series uses its sequence to identify the records. The validation page changes only when an approved run is added.

Installation and calibration consequences

Installation determines the relationship between the sensors, their platform and nearby magnetic sources. Installation and calibration guidance lists, among the items a licensee's installation guidance records, the sensor arrangement and mounting, including orientation and distance from ferrous structures and power equipment. Those details allow a later observation to be related to the installation that produced it.

Mounting away from identified sources is a consideration for the equipment design. A universal separation distance is not supplied here: the relevant configuration includes the particular sensor, structure and electrical equipment. The organisation's evaluation needs to examine the mounting it intends to use. Describing an installation as separated from a source is insufficient unless the record identifies the arrangement and the conditions checked.

The configuration record also identifies the processing unit, power supply, software version, intended environment classes and the set-up sequence. These details distinguish the installed equipment from a general description of geomagnetic positioning. If equipment is replaced or its mounting changes, the subsequent observations need to be associated with the changed configuration. This keeps an earlier acceptance check tied to the arrangement it actually examined.

Checks at the start point

At the start point, the setup record should identify the sensor's calibration status and the magnetic-environment check made before positioning use. Whether calibration is required depends on the sensor. Where its manufacturer provides a procedure, that procedure applies. A requirement to document setup at the start point does not establish that every sensor must be recalibrated at every start. Setup prepares the equipment for work and operation is the period in which it performs its task; Positioning without GPS satellite signals explains that distinction, and the assessment records each input's role and when it is present.

Where calibration forms part of the start-point setup, record that it was performed and the configuration to which it applies. Where it is not required there, retain the applicable calibration status and the pre-use check. This preserves the distinction between the manufacturer's procedure and the preparation of a particular run. It also avoids inventing a common calibration method for different sensor designs.

A change to the sensor, mounting or surrounding structure is recorded as a new configuration. The organisation should identify which checks apply before that configuration enters use. An acceptance check that confirms position output is one part of the installation record; accepting that output for an inspection, display or other task requires the task's assessment criteria as well.

Applications and licensed intellectual property

Geomagnetic positioning supplies position information for a surrounding device, software application or system. Underground, surface, airborne, underwater, indoor and dense urban operation are intended application directions. For disturbance planning, each direction needs a more specific description of the platform, task, electrical activity and magnetic surroundings.

NSATS licenses its intellectual property: patents, designs and copyrighted software. Where the licensed IP takes the form of software, NSATS supplies it under the licence, with its documentation and updates. Each licensee designs and builds its own solution.

The organisation's design connects the sensor installation to the application's use of position. An inspection record, operator display and stationary equipment task can require different observations and acceptance criteria. The disturbance inventory should therefore accompany the application description, so that the evaluation covers the circumstances in which position information will be used. Geomagnetism IP licensing sets out the licensing context.

The executive decision

The decision is whether the proposed implementation has an evaluation plan that covers its own magnetic conditions. Start with the platform and site inventory in Table 1, then identify which equipment states, movements and stationary periods the intended task includes. Each selected condition needs a reference, an observation record and an acceptance criterion appropriate to that task.

The implementation review should address these questions:

- **Platform:** which motors, alternators, steel structures and electrical loads are relevant to the sensor mounting, and which operating states will the evaluation include?
- **Surroundings:** which fixed structures and changing nearby activities belong in the assessment, including stationary work beside cycling machinery?
- **Natural variation:** how will the evaluation record the magnetic conditions relevant to the task without assuming that a space-weather description establishes positioning behaviour?
- **Installation:** which sensor and mounting configuration is being accepted, what calibration procedure applies, and what changes require a new configuration record?
- **Evidence:** who is the assessment owner, who will check the independent reference, how stationary intervals are read, and the relationship between observations and the required position output?

The acceptance decision should identify the conditions covered by the evidence and the conditions still requiring evaluation. The organisation's next assessment can then address a named gap, such as stationary operation under changing electrical loads, rather than treating the general expectation for filtering as a result. How positioning is evaluated provides the framework for documenting that assessment.

Scientific background references

These sources explain natural magnetic fields and variation. They provide scientific background, not evidence of NSATS's filtering or positioning performance.

1. NOAA National Centers for Environmental Information. *International Geomagnetic Reference Field (IGRF)*. Maintained by the International Association of Geomagnetism and Aeronomy. ncei.noaa.gov/products/international-geomagnetic-reference-field.

2. British Geological Survey. *The Earth's Magnetic Field: An Overview*. Sections on field variations at quiet and at disturbed times. geomag.bgs.ac.uk/education/earthmag.html.
3. NOAA Space Weather Prediction Center. *Geomagnetic Storms*. spaceweather.gov/phenomena/geomagnetic-storms.